

Predicting Customer Subscription to a Retail Term Deposit Following a Direct Marketing Campaign Using Machine Learning

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1. Problem Identification

Retail banks offer a range of financial products to meet customers' saving and investment needs. One such product that banks offer to meet those needs is a term deposit, a type of savings account in which customers deposit their money for a fixed period of time, ranging from one month to five years in return for a guaranteed interest rate that is typically higher than that offered by standard savings accounts.

Banks used direct marketing campaigns to promote these products to existing customers. However, marketing every customer was costly and often resulted in low response rates. Identifying customers who were more likely to subscribe could help banks improve the effectiveness of their marketing campaigns and make better use of their resources.

Predicting which customers are more likely to subscribe to a term deposit following a direct marketing campaign represents a suitable machine learning classification problem. Developing a predictive model could help banks target customers more effectively and improve the efficiency of future marketing campaigns. For these reasons, predicting customer subscription to a term deposit following a direct marketing campaign is a suitable business problem for applying machine learning techniques. Therefore, the aim of this project is to develop a machine learning classification model to predict whether a customer will subscribe to a term deposit following a direct marketing campaign.

2. Dataset Selection and Exploration

2.1 Data Source

Dataset	Bank Marketing
Source	UCI Machine Learning Repository
DOI	10.24432/C5K306
File Name	bank-additional-full.csv

The dataset used in this project was obtained from the UCI Machine Learning Repository. The Bank Marketing dataset was used because it aligns with the business problem identified in Section 1: predicting whether a customer will subscribe to a term deposit following a direct telephone marketing campaign.

The file, bank-additional-full.csv was chosen because this is the full version of dataset and contains 41,188 customer records with five new additional economic and social variables. The data make it suitable for developing a binary classification model. This dataset is also almost identical to the one used by a real business to predict success of bank telemarketing by a data-driven approach.

2.2 Data Description

The dataset contains customer and campaign information collected during direct telephone marketing campaigns conducted by a Portuguese banking institution. In many cases, customers were contacted more than once before a final outcome was recorded.

The objective of the dataset is to predict whether a customer will subscribe to a term deposit following a telephone marketing campaign. This makes the problem a binary classification task, where the target variable (y) has two possible outcomes: yes (subscribed) or no (did not subscribe).

The dataset contains 41,188 customer records and 20 input variables covering customer demographics, financial information, campaign history and macroeconomic indicators together with one target variable indicating whether the customer subscribed to a term deposit.

Attribute	Value
Number of records	41,188
Input variables	20
Target variable	y
Prediction type	Binary classification
Positive class	yes
Negative class	no

2.3 Data Limitation and Project Scope

This project uses the UCI Bank Marketing dataset, which contains historical data from telephone-based direct marketing campaigns conducted by a Portuguese banking institution. Consequently, the scope of this project is limited to predicting customer subscription to a term deposit following telephone marketing campaigns.

Although modern banks increasingly use multichannel marketing approaches, such as email, mobile applications and online banking platforms, these channels are not represented in the dataset and therefore fall outside of the scope of this project. As a result, the predictive model developed in this project should only be used in relation to telephone-based marketing campaigns rather than modern multichannel marketing strategies, as the result could be significantly different as different demographics use different media methods.

The dataset document also stated that several categorical variables contain missing values represented by the category 'unknown'. The appropriate treatment of these values will be determined during the data preprocessing stage where they will be assessed to decide whether they should be retained as a valid category or handled using deletion or imputation techniques.

3. Data Splitting

The Data Frame was split into training (80%) and testing (20%) sets using stratified sampling to preserve the class distribution, while the test set was reserved for the final evaluation of model performance. This approach minimises the risk of data leakage and provides an unbiased

estimate of the model's ability to generalise to unseen data. At this stage, X (input variables) and y (target) were not separated as having complete training dataset helps to perform better data preprocessing.

Only the training dataset was profiled and preprocessed in this assignment. The test dataset was not used during preprocessing to prevent data leakage, ensuring that information from the test set did not influence preprocessing decisions. The test dataset was reserved for the final evaluation of the machine learning models in the next stage of the project.

4. Data Preprocessing

4.1 Data Profiling

The training dataset was profiled by examining the following:

- Data size and structure
- Content overview
- Missing values / 'unknown' values
- Duplicate records
- Format inconsistency (Whitespace)

The training set consists of 32,950 rows and 21 columns with the 3 data types of int64, str and float64 with no missing values, although the documents stated that there are several categorical variables containing missing values represented by 'unknown'. The 6 categories which contain 'unknown' value were Job, Marital, Education, Default, Housing and Loan.

Each category contains the below categorical values.

Categorical Variable	Unique value count	Categorical values
Job	12	'blue-collar', 'entrepreneur', 'retired', 'admin.', 'student', 'services', 'technician', 'self-employed', 'management', 'unemployed', 'unknown', 'housemaid'
Marital	4	'married', 'divorced', 'single', 'unknown'
Education	8	'basic.9y', 'university.degree', 'basic.4y', 'high.school', 'professional.course', 'unknown', 'basic.6y', 'illiterate'
Default	3	'unknown', 'no', 'yes'
Housing	3	'no', 'yes', 'unknown'
Loan	3	'no', 'yes', 'unknown'
Contact	2	'cellular', 'telephone'
Month	10	'nov', 'jul', 'may', 'jun', 'aug', 'mar', 'oct', 'apr', 'sep', 'dec'
Days of week	5	'wed', 'mon', 'tue', 'fri', 'thu'
Poutcome	3	'nonexistent', 'failure', 'success'
Y	2	'no', 'yes'

4.2 Preprocessing decisions

There are a several types of contents that then needed specific treatments to be decided:

- 'Unknown' values
- Duplicated records

- Encoding
- Class imbalance

‘Unknown’ values:

The treatment of ‘unknown’ values was determined on a feature-by-feature basis rather than applying a single imputation strategy to all categorical variables. For the housing and loan variables, the proportion of ‘unknown’ values was low and the subscription rates of customers with ‘unknown’ values were almost identical to those of the observed categories. Therefore, it was decided to replace missing values with the most frequent category as it was considered unlikely to introduce meaningful bias.

In contrast, for the education variable it was decided to retain the ‘unknown’ category because mode imputation would have substantially altered the original category distribution, even though its overall percentage against the whole training dataset was less than 5%.

‘Unknown’ values proportion in each category is shown in the below table:

Variable	‘Unknown’ value count	‘Unknown’ count percentage
Job	265	0.80%
Marital	65	0.20%
Education	1,397	4.24%
Default	6,940	21.06%
Housing	796	2.42%
Loan	796	2.42%

The decisions were made as below:

Variable	Decision	Reason and Treatment
Job	Imputation	Imputation method was applied with the most frequent value, ‘admin.’ as the impact will be minimum using SimpleImputer(strategy=‘most_frequent’).
Marital	Imputation	Imputation method was applied with the most frequent value, ‘married’ as the impact will be minimum using SimpleImputer(strategy=‘most_frequent’).
Education	Keep as ‘unknown’	Even though the ‘unknown’ values in education is less than 5%, if the most frequent value applied, university.degree appearance goes up by 14%. This will potentially distort the original distribution about customers’ education level. Therefore the ‘unknown’ values in education was retained to avoid introducing unsupported assumptions about customers’ education levels.
Default	Keep as ‘unknown’	The default = no and target = yes category contains 12.9% within the whole the default = no category, while the unknown category with target = yes showed a substantially lower subscription rate (5.2%) within the whole unknown category. This indicates that the unknown values contain useful information. Therefore the ‘unknown’ values were

		retained as a separate category rather than being replaced with the most frequent value.
Housing	Imputation	The 'unknown' values were in the same rows for both Housing and Loan columns. This indicates that the bank did not record housing and loan information for the same customers. Imputation method was still be applied with the most frequent values, 'yes' as the impact will be minimum. This is based on the investigation, whether 'unknown' have almost exactly the same subscription rate as the majority.
Loan	Imputation	With the same reason as Housing columns, the imputation method was applied with the most frequent values, 'no'

Duplicated records:

Eight completely duplicated records were identified in the training dataset. Although these could represent data entry errors, the dataset does not contain a unique customer identifier or timestamp to determine whether they are accidental duplicates or legitimate repeated observations from separate telephone contacts. Given that they represent only approximately 0.02% of the training data, they were retained to avoid removing potentially valid observations.

Encoding:

Encoding was performed only for categorical variables excluding y (target) leaving it 'yes'/'no' because algorithms only process numerical data when it comes to machine learning.

Initially `pd.get_dummies()` was considered for encoding the categorical variables. However, after further research about one hot encoding, `OneHotEncoder` from `scikit-learn` was selected because it can handle unseen categories in new data and integrates more effectively into machine learning pipelines.

Class imbalance:

The target variable was investigated for class balance. Approximately 88.7% of customers did not subscribe to a term deposit while 11.3% did subscribe. The train-test split was used with a stratified parameter resulting in almost identical class distributions across the training and test datasets. The class imbalance will be considered during the model development.

Combine imputed data frame and encoded data frame:

After imputing the 'unknown' values in the Job, Marital, Housing, Loan variables and encoded categorical columns (excluding the target column), the two data frames were combined using the `concat()` function. The encoded data frame created a separate binary column for each categorical value, so these new columns were added while the original categorical columns were removed. This ensured that the final clean dataset contained only numerical features required for training machine learning models.

5. Exploratory Data Analysis

5.1 Target variable distribution

The target variable was imbalanced with a considerably higher proportion of customers with no subscription to a term deposit. The imbalance was investigated during preprocessing but was not addressed at this stage because no machine learning models were trained in the assignment.

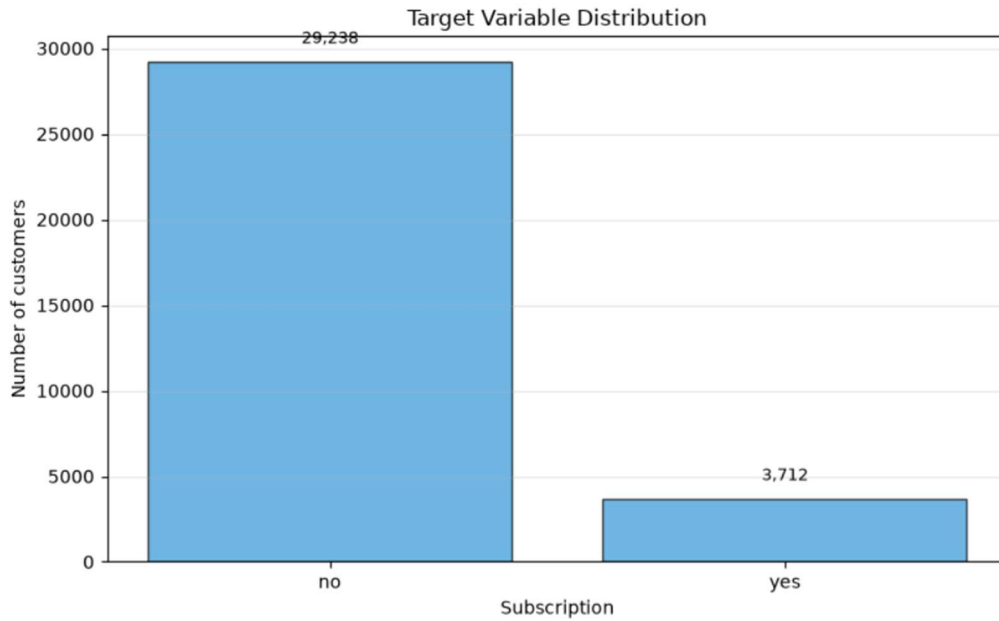


Figure 5.1: Target Variable Distribution

5.2 Age Distribution

The histogram shows that customers' ages range from 17 years old to 98 years old with the highest concentration between 30 to 40 years old. The distribution is slightly right-skewed, indicating that relatively few customers are in the older age groups.

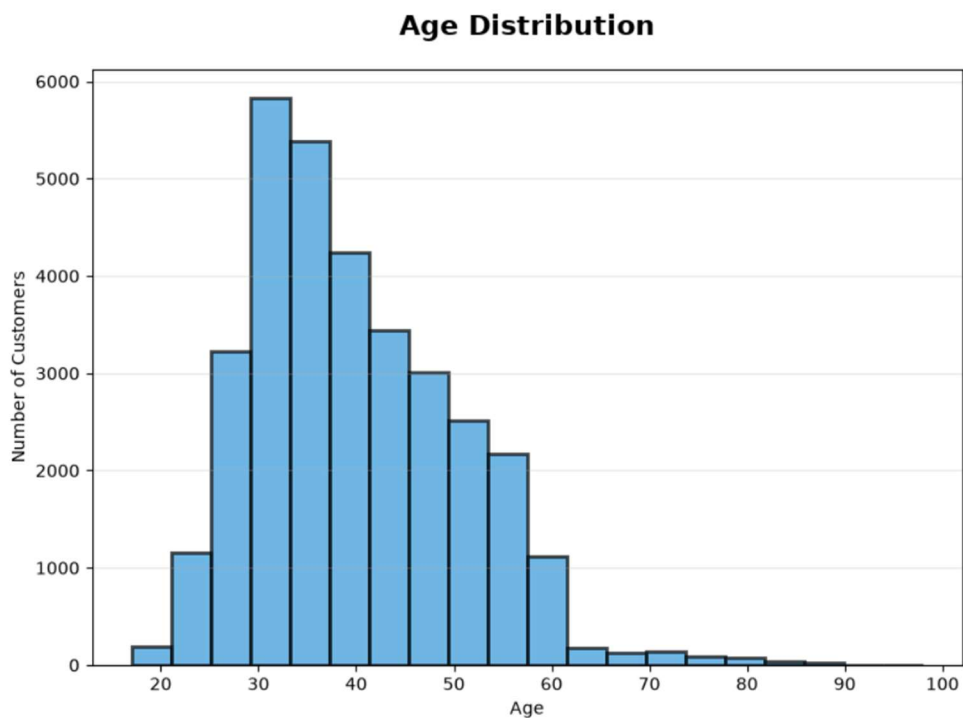


Figure 5.2: Age Distribution

5.3 Duration vs Subscription

The boxplot shows the relationship between the call duration and subscription to a term deposit results and those who did not subscribe. Customers who subscribed generally had a longer call durations as indicated by the higher median and wider interquartile range. Both groups contain significant outliers, suggesting that some telephone calls lasted substantially longer than most calls.

As stated in the documentation, there is a limitation in duration records that logically call duration is only known *after* the telephone conversation has ended. Although it appears to be associated with term deposit subscription, it would not be available when predicting customer subscription before a marketing call. Therefore its practical use in a prediction model is limited even though it provides useful insight during EDA.

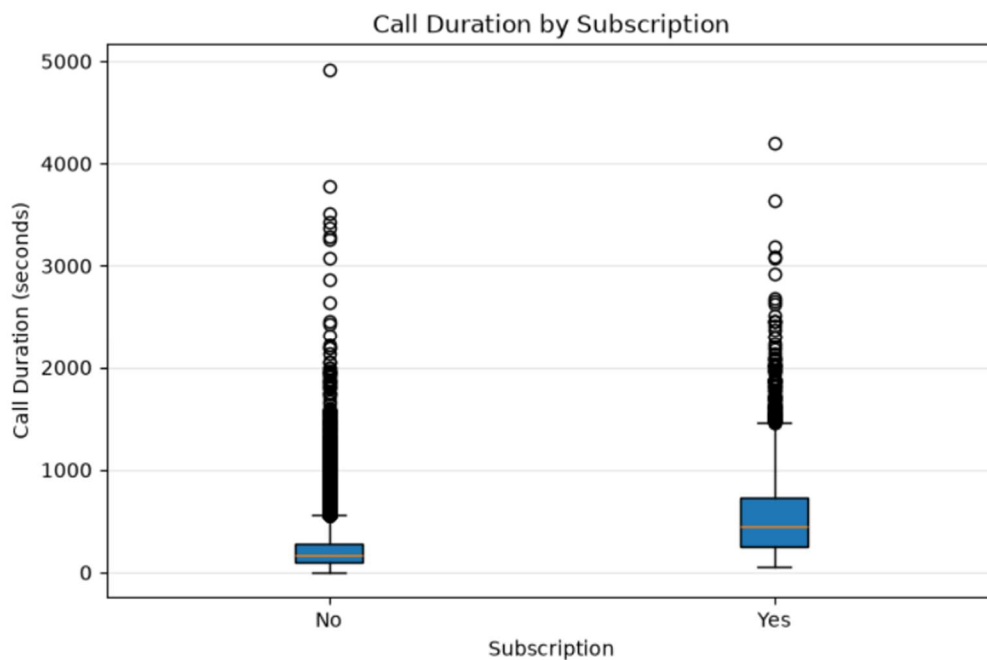


Figure 5.3: Call Duration by Subscription

5.4 Subscription Rate by Job

The bar chart of subscription rate by job shows that subscription rates vary across job categories. Within the training dataset, students and retired customers have the highest subscription rates whereas blue-collar workers have the lowest. This variation suggests that occupation may provide useful predictive information for predicting customer subscription when training machine learning models.

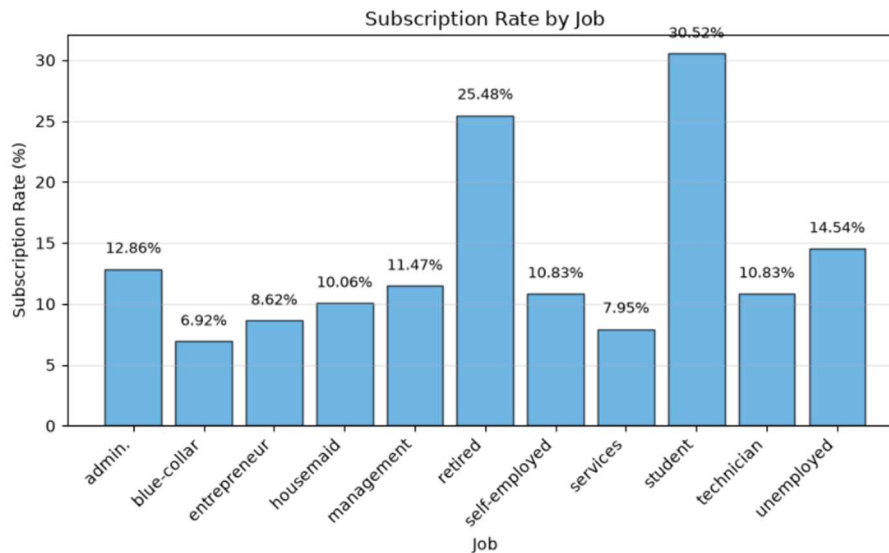


Figure 5.4: Subscription Rate by Job

5.5 Subscription Rate by Education

The bar chart shows that the subscription rate across different education levels in the training dataset. Subscription rates vary between education categories. Customers with a university degree (13.81%) and those with an unknown education level (14.82%) had relatively higher subscription rates than customers with basic education levels. Although the illiterate category had the highest subscription rate (18.75%), this category contained only 16 customers (3 subscribed and 13 did not). Therefore this result should be interpreted with caution because it is based on a very small sample. The relatively high subscription rate for the unknown education category also supports the preprocessing decision to retain 'unknown' as a separate category rather than replacing it with the most frequent value. Overall, the variation across education levels suggests that education may provide useful information for predicting term deposit subscription.

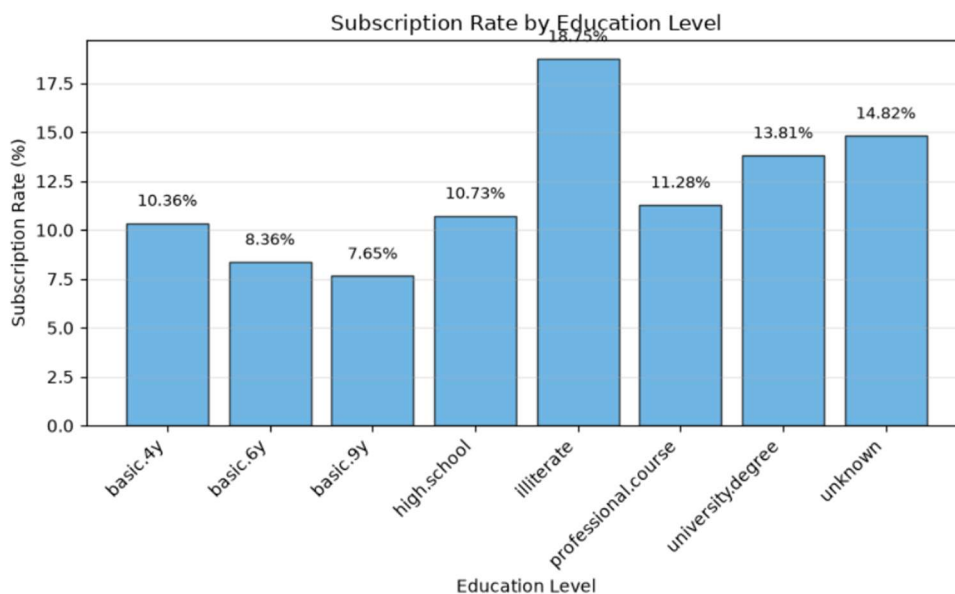


Figure 5.5: Subscription Rate by Education

6. Further Analysis Before Preprocessing the Whole Training Dataset

As potential outliers were identified in the EDA section and scaling was required for model training, further analysis of the numerical variables was carried out.

6.1 Further Analysis on the Numerical Variables

The boxplots identified potential outliers in Age and Campaign while Pdays and Previous showed highly concentrated distributions. Further analysis was therefore carried out on Age, Campaign, Pdays and Previous using histograms to examine their distributions and identify any additional patterns.

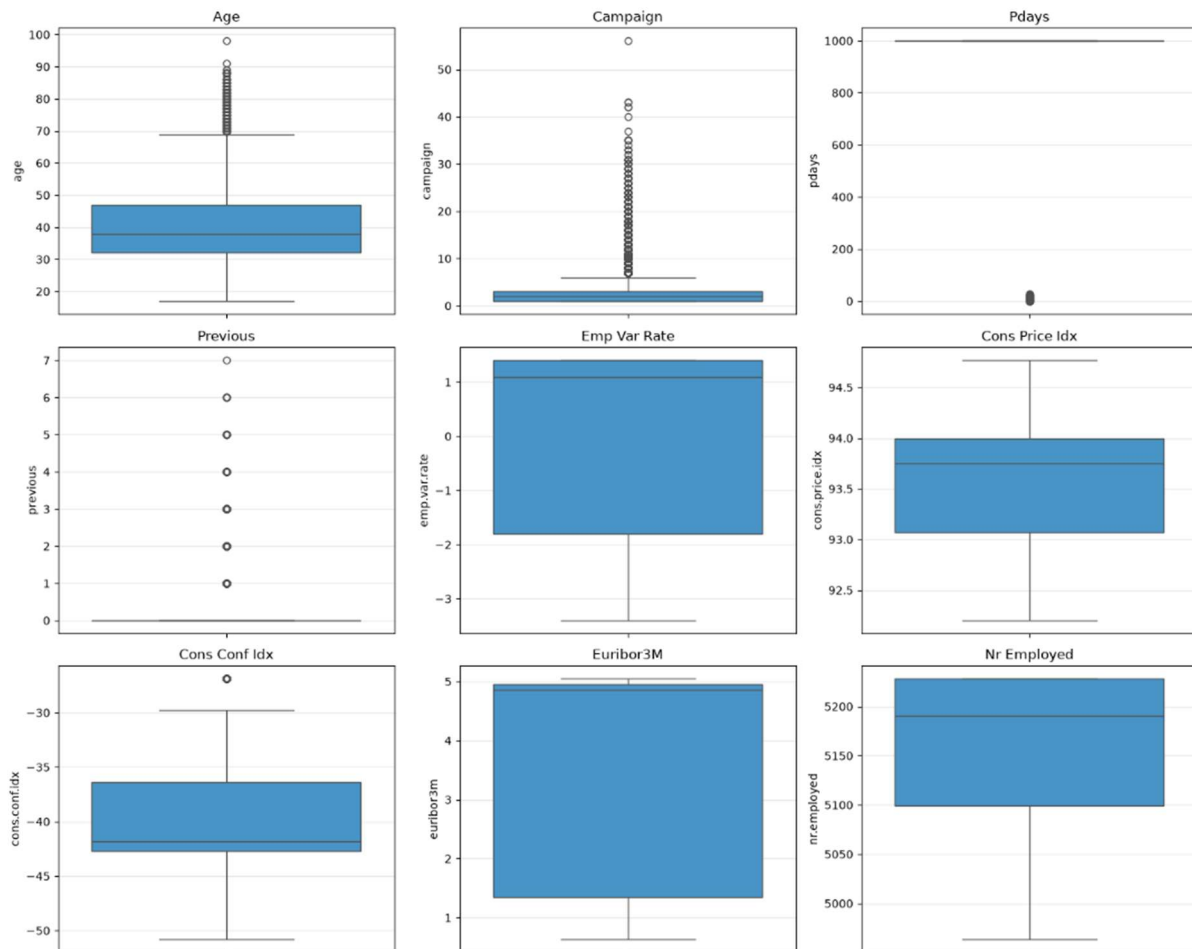


Figure 6.1: Boxplots of the Numerical Variables in the Training Dataset

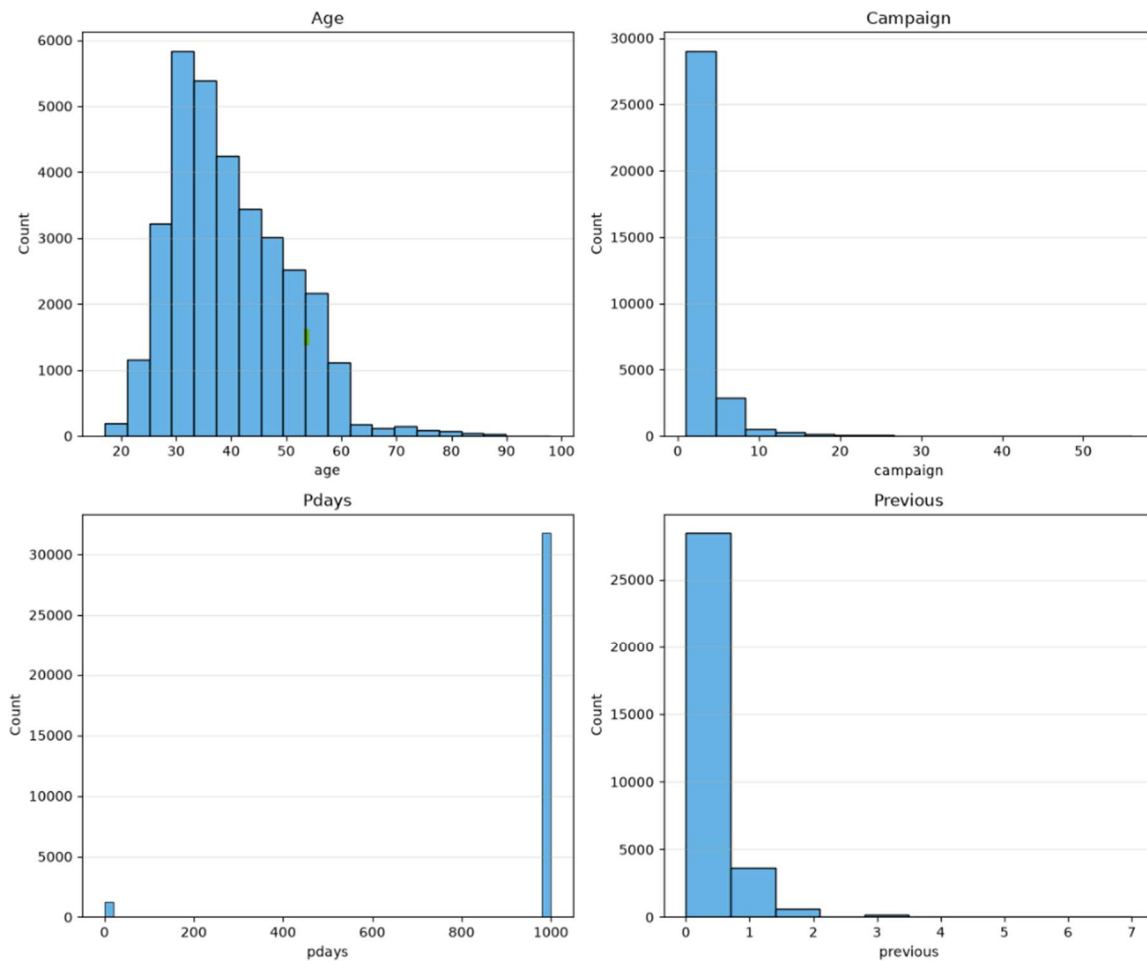


Figure 6.2: Histograms of Age, Campaign, Pdays and Previous in the Training Dataset

Variable	Finding	Reason and Treatment
Age	The variable exhibited a slightly <i>right-skewed</i> distribution. The statistical summary showed ages ranging from 17 to 98 years with a median age of 38 years. The boxplot identified some potential outliers in the upper age range.	The observed age range (17-98 years) was reasonable with no invalid age values identified. Therefore, all observations were retained.
Campaign	The variable exhibited a highly <i>right-skewed</i> distribution. The median number of contacts was 2 with values ranging from 1 to 56. The boxplot identified values above approximately five contacts as potential outliers using the IQR rule. These observations represented 8.16% of the training dataset.	Although some high values were identified, they represent the number of contacts made during the marketing campaign and may reflect genuine customer interactions. Therefore, all observations were retained.
Pdays	The variable was highly concentrated at the value 999 as shown in the histogram.	The value 999 is a coded value indicating that a customer was not previously contacted. Therefore, these values were retained, and further considered during feature engineering.

Previous	The variable was highly concentrated at 0 as shown in the histogram, indicating that most customers had not been contacted in previous marketing campaigns.	The distribution reflects customers' previous marketing campaign history. Therefore, all observations were retained.
Emp.var.rate	The variable included both positive and negative values reflecting periods of increasing employment (positive) and decreasing employment (negative). As a quarterly indicator, the values represent changes in employment levels during a given period rather than individual customer characteristics. No unusual values were identified in the boxplot.	The values indicate employment conditions during the marketing campaign and were considered valid. Therefore, all observations were retained.
Cons.price.idx	The Consumer Price Index variable showed a relatively narrow distribution with no unusual values identified in the boxplot. As a monthly indicator, the values represent consumer price levels during a given period rather than individual customer characteristics.	The values indicate consumer price conditions during the marketing campaign and were considered valid. Therefore, all observations were retained.
Cons.conf.idx	The Consumer Confidence Index variable contained only negative values, indicating relatively low consumer confidence during the marketing campaign period. No unusual values were identified in the boxplot.	The values indicate consumer confidence during the marketing campaign and were considered valid. Therefore, all observations were retained.
Euribor3m	The 3-month Euro Interbank Offered Rate variable showed variation reflecting changes in market interest rates in the Eurozone during the marketing campaign period. No unusual values were identified in the boxplot.	The values indicate interest rate conditions during the marketing campaign and were considered valid. Therefore, all observations were retained.
Nr.employed	The number of Employees variable showed variation reflecting changes in employment levels during the marketing campaign period. No unusual values were identified in the boxplot.	The values indicate employment numbers during the marketing campaign and were considered valid. Therefore, all observations were retained.

Although some numerical variables contained outliers, all observations were retained because they represented valid values. Therefore, RobustScaler, which uses the median and the interquartile range (IQR) and is less sensitive to outliers, was selected to reduce the influence of extreme values during the model training.

6.2 Correlation Analysis on the Numerical Variables

To understand the relationship between numerical variables, correlation analysis was performed.

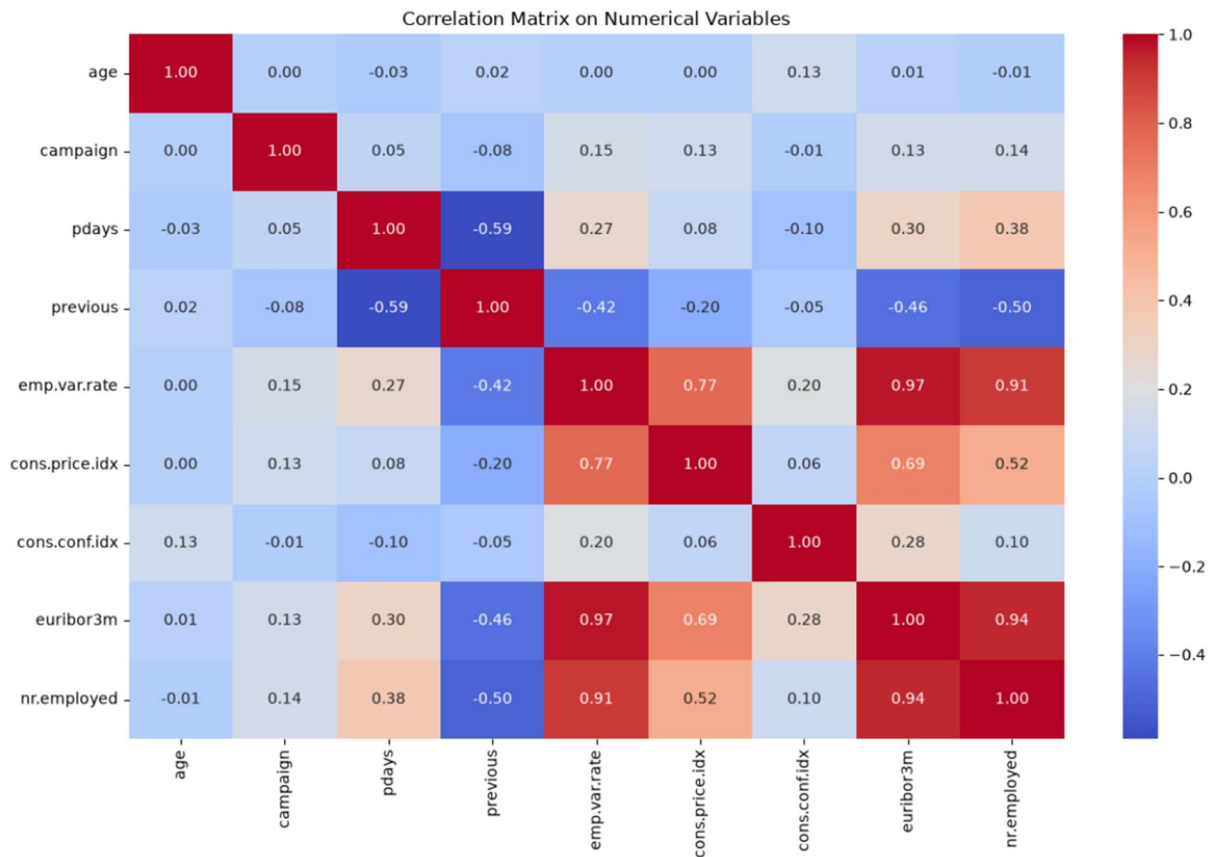


Figure 6.3: Heatmap of the Numerical Variables

The correlation matrix showed that most numerical variables had weak correlations with one another. However, a few macroeconomic indicators showed strong positive correlations. In particular, `emp.var.rate`, `euribor3m`, and `nr.employed` had correlation coefficients above 0.90, suggesting potential multicollinearity. These variables were retained because they represent different economic measures and no variables were removed based on correlation alone.

6.3 Further Analysis of Categorical Variables

Categorical variables were analysed using both customer counts and within-category subscription rates. The percentage plots supported comparison across categories of different sizes, while the count plots were used to identify categories with small sample sizes whose subscription rates could otherwise be misleading.

Variable	Finding
Job	Students and retired customers showed higher subscription rates than the other job categories. However, these groups contained fewer customers than the largest job categories indicating that the higher subscription rates were based on smaller sample sizes.
Marital	Single customers showed the highest subscription rate among the marital categories. The count plot indicates that this observation was not driven by a small sample.
Education	The illiterate category showed the highest subscription rate. However, this category contained only 16 customers, so the result was based on a very small sample. Apart from the illiterate category, customers with a high

	school, a professional course and a university degree showed slightly higher subscription rates than the basic education levels.
Default	The default = yes category had no customer subscriptions. However, this category contained only 3 customers so no meaningful conclusion can be made. Among the remaining categories, customers with default = no showed a higher subscription rate than those with default = unknown.
Housing	Customers with and without housing loans showed similar subscription rates of approximately 11% suggesting little association between housing loan status and subscription.
Loan	Personal loan status showed a similar pattern with housing loan status. Customers with and without personal loans showed similar subscription rates of approximately 11%, suggesting little association between personal loan status and subscription.
Contact	Customers contacted by cellular phone had higher subscription rates than those contacted by telephone.
Month	Months with fewer customer contacts such as March, September, October, December showed relatively higher subscription rates than the months with larger customer counts.
Day of Week	Subscription rates were similar across all days of the week suggesting little association between the day of contact and customer subscription.
Poutcome	Although the number of customers with a previous successful campaign outcome was relatively small, this group showed the highest subscription rate. This suggests that customers who responded successfully to a previous marketing campaign were more likely to subscribe again.

7. Feature and Target Preparation

The dataset was separated into features (X) and the target variable (y). The target variable was then converted into binary values with 'no' = 0 and 'yes' = 1 for model training and evaluation.

8. Test Data Preprocessing

The same imputation and OneHotEncoding methods fitted on the training data were applied to the test data. The preprocessing methods were not refitted on the test data to prevent data leakage and ensure that the test set remained unseen during model development.

One-hot encoding retained all categories rather than dropping a reference category. Dropping a reference category would not remove those observations from the dataset, instead, that category would be represented when all other category columns are zero. All categories were retained in this project to maintain a consistent representation across the selected models.

9. Feature Engineering

Further analysis in Section 6 showed that the pdays variable was highly concentrated at 999 with 96% of customers having this value.

Since 999 indicates that a customer was not contacted in a previous marketing campaign rather than representing the actual number of days since the last contact, pdays was converted into a binary feature (pcontacted). Furthermore, the analysis of the poutcome variable showed that customers who had responded successfully to a previous marketing campaign were more likely to subscribe to a term deposit. Therefore, representing previous contact as a binary feature was considered to better reflect the underlying meaning of pdays and provide useful information for model training.

The new feature, pcontacted, was coded as 1 for customers who had been contacted previously and 0 for customers who had not been contacted previously.

The original pdays variable was removed.

10. Feature Scaling

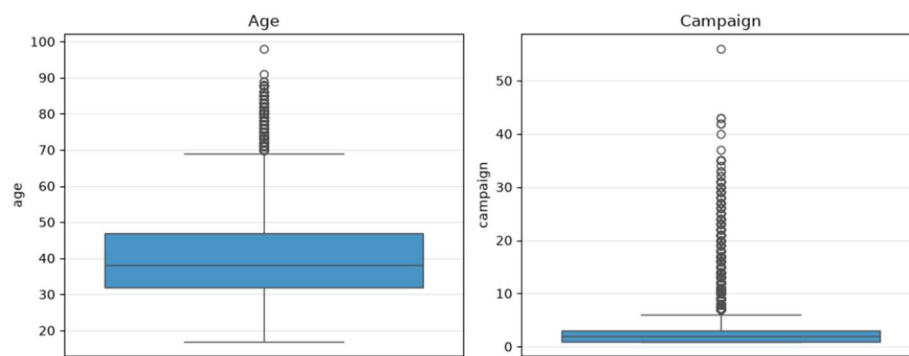
Based on the further analysis of the numerical variables, RobustScaler was selected because some numerical variables contained outliers, while all observations were retained because they represented valid values.

RobustScaler uses the median and the interquartile range (IQR), making it less sensitive to outliers and reducing the influence of extreme values during model training.

As shown in the boxplots below, the outliers in Age and Campaign were retained after scaling because they represented valid observations. The overall distributions remained unchanged, while RobustScaler centred each variable around its median and rescaled it using the interquartile range (IQR). This reduced the influence of extreme values without removing any observations.

The scaler was fitted on the training data and then applied to the test data without refitting to prevent data leakage.

Age and Campaign Boxplots Before Scaling:



Age and Campaign Boxplots After Scaling:

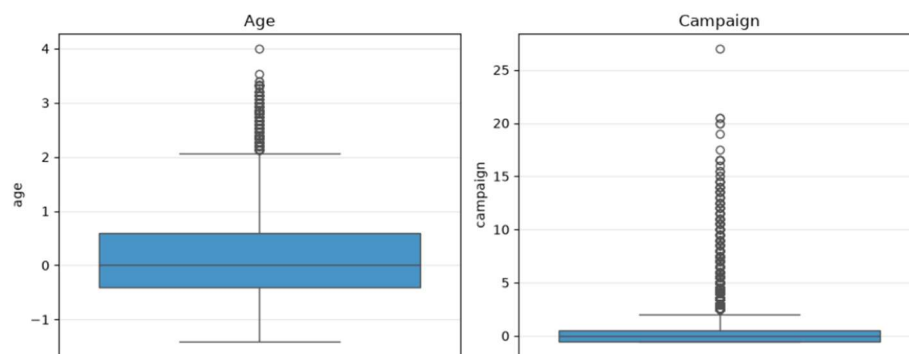


Figure 10.1: Comparison of Age and Campaign Before and After Scaling

11. Model Selection and Experimentation

Three classification models were selected for experimentation in this project.

Model	Selection reason
Logistic Regression	The model was selected because it is a simple and interpretable classification algorithm suitable for predicting a binary outcome.
Support Vector Machines (SVM)	SVM was selected because it is suitable for classification with multiple features and can identify a decision boundary that maximises the separation between classes. SVC was used as the classification implementation of SVM.
Random Forest	The model was selected because it can capture non-linear relationships and interactions between features by combining multiple decision trees.

11.1 Baseline Model Comparison

As mentioned in the EDA section, the target was imbalanced with approximately 89% of customers not subscribing to a term deposit and 11% subscribing.

The baseline models were trained on the original imbalanced training data before tuning hyperparameters.

The baseline models were evaluated using 5-fold cross-validation on the training data, keeping the test dataset untouched for the final model evaluation.

F1 score was used as the evaluation metric because the target variable was imbalanced and F1 provides a balance between precision and recall.

Baseline Model	Mean CV F1 Score
Logistic Regression	0.336
SVC	0.321
Random Forest	0.358

Conclusion:

Random Forest achieved the highest mean CV F1 score of 0.358 among the baseline models.

12. Hyperparameter Tuning

Hyperparameter tuning was performed to improve the three baseline models. GridSearchCV with 5-fold cross-validation and F1 score was used for Logistic Regression and Random Forest. For SVC, GridSearchCV was initially attempted but manual tuning was used due to the computational time required. The test dataset remained untouched throughout the tuning process and was reserved for final model evaluation.

12.1 Logistic Regression

Hyperparameter	Value Tested	Reason
C	0.01, 0.1, 1, 10	Values were selected on a logarithmic scale around the default value of 1 to test a range of regularisation strengths. Smaller C values apply stronger regularisation while larger C values apply weaker regularisation. 0.01: Strong regularisation 0.1: Moderately strong regularisation 1: Default / middle reference point 10: Weaker regularisation
Class_weight	None, balanced	As the training dataset is class imbalanced, 'None' and 'balanced' were tested to evaluate whether adjusting class weights improves the model's ability to identify the minority class.

Initial tuning:

Best parameters: C = 10, class_weight = 'balanced'

Mean cross-validation F1 score: 0.451

Refined tuning:

As the initial tuning selected C = 10 which provided the weakest regularisation among the values tested. Based on the result, further tuning was performed with C = 10 and C = 100 to investigate whether weaker regularisation could improve model performance while retaining class_weight = 'balanced'.

Conclusion:

The refined tuning still selected C = 10 with class_weight = 'balanced'. Therefore, the parameters from the initial tuning were retained.

12.2 SVC**Initial tuning approach:**

GridSearchCV was initially attempted to test combinations of C, class_weight and kernel using 5-fold cross-validation and F1 score. However, the process was not computationally feasible to complete in either VS Code or Google Colab.

Hyperparameter	Value Tested	Reason
C	0.01, 0.1, 1, 10	Different C values were tested to control the trade-off between maximising the margin and reducing classification errors. Values were selected on a logarithmic scale around the default value of it. 0.01: Strong regularisation 0.1: Moderately strong regularisation 1: Default / middle reference point 10: Weaker regularisation
Class_weight	None, balanced	As the training dataset is class imbalanced, 'None' and 'balanced' were tested to evaluate whether adjusting class weights improves the model's ability to identify the minority class.

Kernel	Linear, RBF	Linear and RBF kernels were selected to compare linear and non-linear decision boundaries while keeping the hyperparameter search computationally manageable.
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SVC tuning:

Therefore, two SVC models were tested manually using 5-fold cross-validation:

- Manual tuning 1: class_weight = 'balanced' and kernel = 'rbf'
- Manual tuning 2: class_weight = 'balanced' and kernel = 'linear'

For both models, the C value was kept at its default value of 1.

SVC Tuning	Kernel	Mean CV F1 Score
Manual tuning 1	RBF	0.468
Manual tuning 2	Linear	0.452

Conclusion:

Manual tuning 1 with the RBF kernel achieved the higher mean CV F1 score of 0.468 compared with 0.452 for the linear kernel. Therefore, Manual tuning 1 was selected as the tuned SVC model.

12.3 Random Forest

Hyperparameter	Value Tested	Reason
N_estimators	100, 200	100: Default (same as baseline model) 200: Tests whether increasing the number of trees improves model performance and stability.
Max_depth	None, 10, 20	The default unrestricted depth was compared with depths of 10 and 20 to test whether controlling tree complexity improves model performance and reduces overfitting. None: Default (same as baseline model), with no specified maximum depth. 10: shallower trees 20: deeper trees
Min_samples_split	2, 5	The default value of 2 was compared with 5 to test whether requiring more samples to split a node helps control tree complexity and reduce overfitting.
Class_weight	None, balanced	To evaluate whether adjusting class weights improves the model's ability to identify the minority class.

Initial tuning:

Best parameters: class_weight = 'balanced', max_depth = 20, min_samples_split = 5, n_estimators = 200

Mean cross-validation F1 score: 0.4803

Refined tuning:

As the initial tuning selected `max_depth = 20`, `min_samples_split = 5`, `n_estimators = 200`, which were at the upper end of the numerical values tested, further tuning was performed. Based on the result, `n_estimators = 200` and `300`, `max_depth = 20` and `30`, `min_samples_split = 5` and `10` were tested while retaining `class_weight = 'balanced'` to investigate whether increasing these values improved model performance.

Refined tuning result:

Best parameters: `class_weight = 'balanced'`, `max_depth = 20`, `min_samples_split = 10`, `n_estimators = 300`

Mean cross-validation F1 score: 0.4811

Conclusion:

The refined tuning achieved a slightly higher mean CV F1 score of 0.4811 compared with 0.4803 from the initial tuning.

The improvement was very small, so it was decided that no further model refinement was required for Random Forest.

Therefore, the refined parameters were selected for the tuned Random Forest model.

13. Overall Model Comparison

The mean CV F1 scores of each baseline model was compared with its final turned version. The best performing model was then selected for final evaluation.

Model	Mean CV F1 Score
Baseline Logistic Regression	0.335635
Tuned Logistic Regression	0.450754
Baseline SVC	0.321003
Tuned SVC (RBF)	0.468371
Baseline Random Forest	0.358447
Refined Random Forest	0.481107

All three models achieved higher mean CV F1 scores after tuning compared with their baseline models.

Refined Random Forest achieved the highest mean CV F1 score of 0.4811. Therefore, the refined Random Forest was selected as the final model for evaluation on the test dataset.

14. Final Model Evaluation on the Test Dataset

The refined Random Forest model was selected based on the highest mean cross-validation F1 score.

The selected model was evaluated on the untouched test dataset to assess its performance on unseen data.

	Precision	Recall	F1-score	Support
not_subscribed	0.949108	0.895486	0.921518	7,310
subscribed	0.430276	0.621767	0.508594	928
accuracy	0.864652	0.864652	0.864652	0.864652
macro avg	0.689692	0.758626	0.715056	8,238
weighted avg	0.890663	0.864652	0.875002	8,238

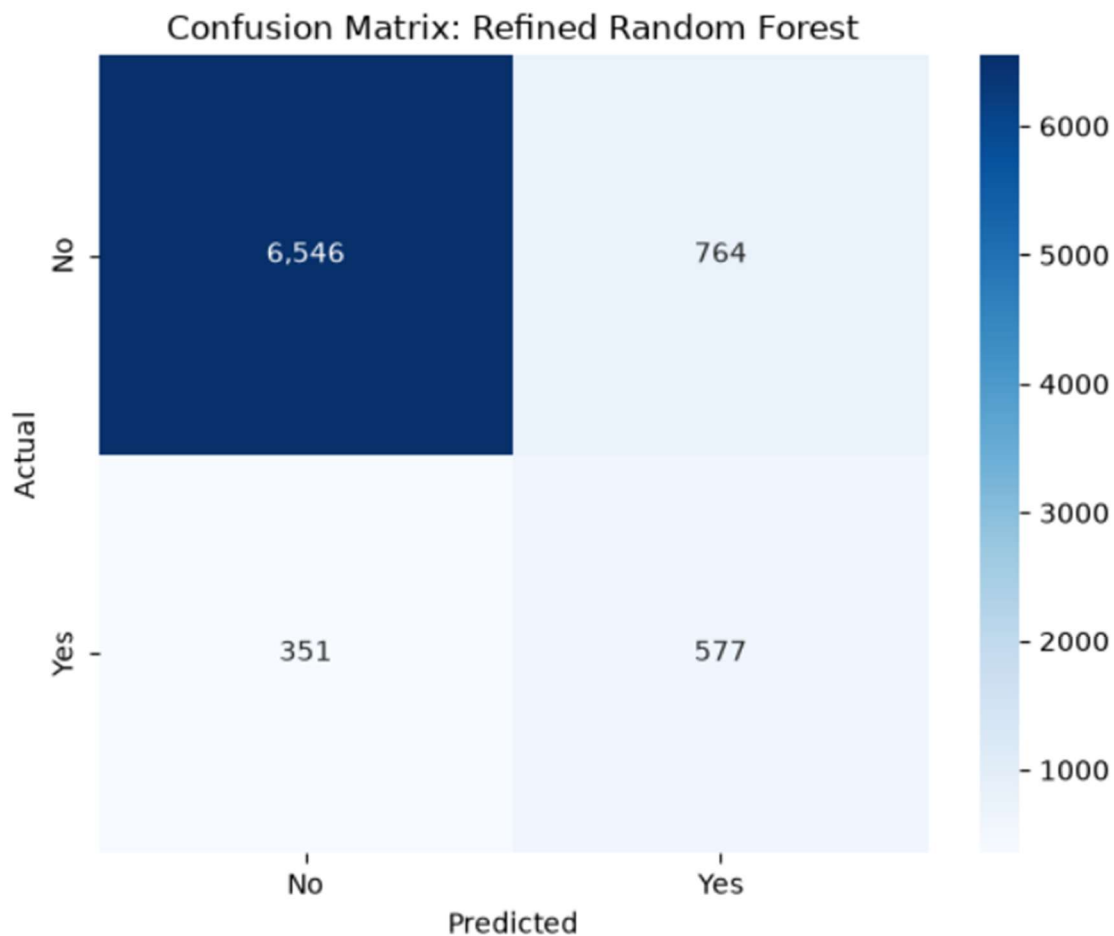


Figure 14.1: Confusion Matrix for the Refined Random Forest

The refined Random Forest achieved an F1 score of 0.509 for subscribed customers on the unseen test dataset, with precision of 0.430 and recall of 0.622. The confusion matrix also supported these results, showing that the model correctly identified 577 out of 928 customers who subscribed, while 351 subscribers were not identified. There were also 764 customers incorrectly predicted as subscribers.

Although the overall accuracy was 86.5%, the F1 score, precision and recall for the subscribed class provide a more meaningful evaluation due to the class imbalance. Recall is particularly relevant to the business objective because the model aims to identify customers who are more likely to subscribe. The recall of 0.622 indicates that the model identified 62.2% of actual subscribers. However, the precision of 0.430 also shows that some customers predicted as subscribers did not subscribe, which could result in unnecessary marketing contacts.

15. Conclusion

Three classification models were evaluated and improved through hyperparameter tuning and class weighting. All three models achieved higher mean CV F1 scores after tuning, with the Refined Random Forest achieving the highest score of 0.4811 and therefore being selected as the final model.

The final evaluation showed that the Refined Random Forest identified 62.2% of actual subscribers on the unseen test dataset, with an F1 score of 0.509 for the subscribed class. However, the precision of 0.430 showed that some customers predicted as subscribers did not subscribe. While the model demonstrated some ability to identify potential subscribers, its overall performance for the subscribed class was limited and further improvement would be required before it could be considered effective for practical marketing decisions.

Further development could investigate alternative approaches to class imbalance, such as SMOTE, and use a pipeline to integrate preprocessing and resampling within cross-validation.

The impact of the highly correlated macroeconomic variables could also be investigated by comparing model performance with and without selected correlated features.

Future development could also investigate whether dropping a reference category during one-hot encoding affects model performance, particularly for Logistic Regression.

Overall, this project demonstrated the process of developing and evaluating classification models for predicting term deposit subscription. Although the Refined Random Forest achieved the strongest performance among the models tested, the final results showed that further development is required to improve its ability to identify potential subscribers accurately. As a future extension of this project, I would continue experimenting with class imbalance techniques, feature selection and preprocessing approaches to investigate whether the model's performance could be improved.